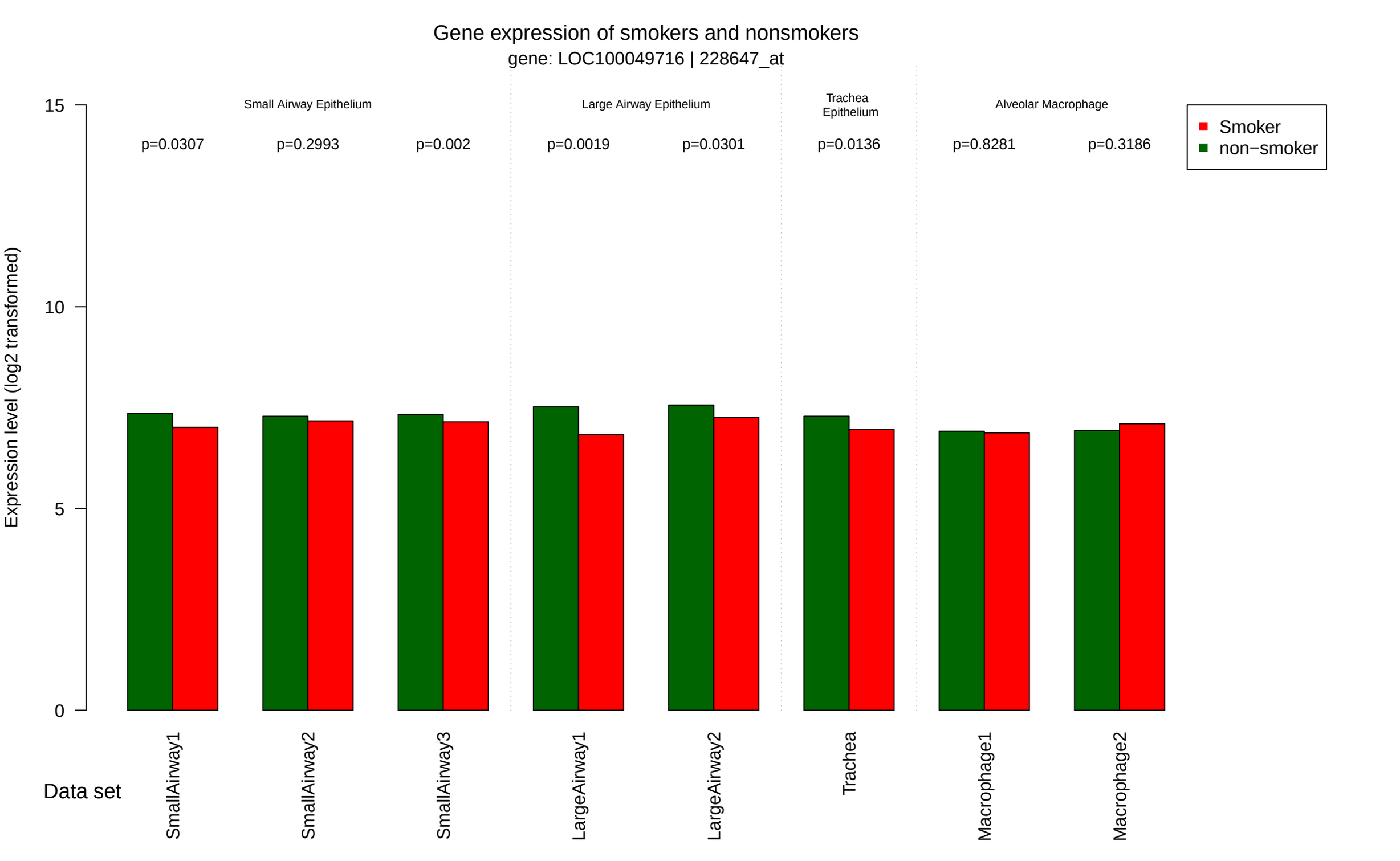
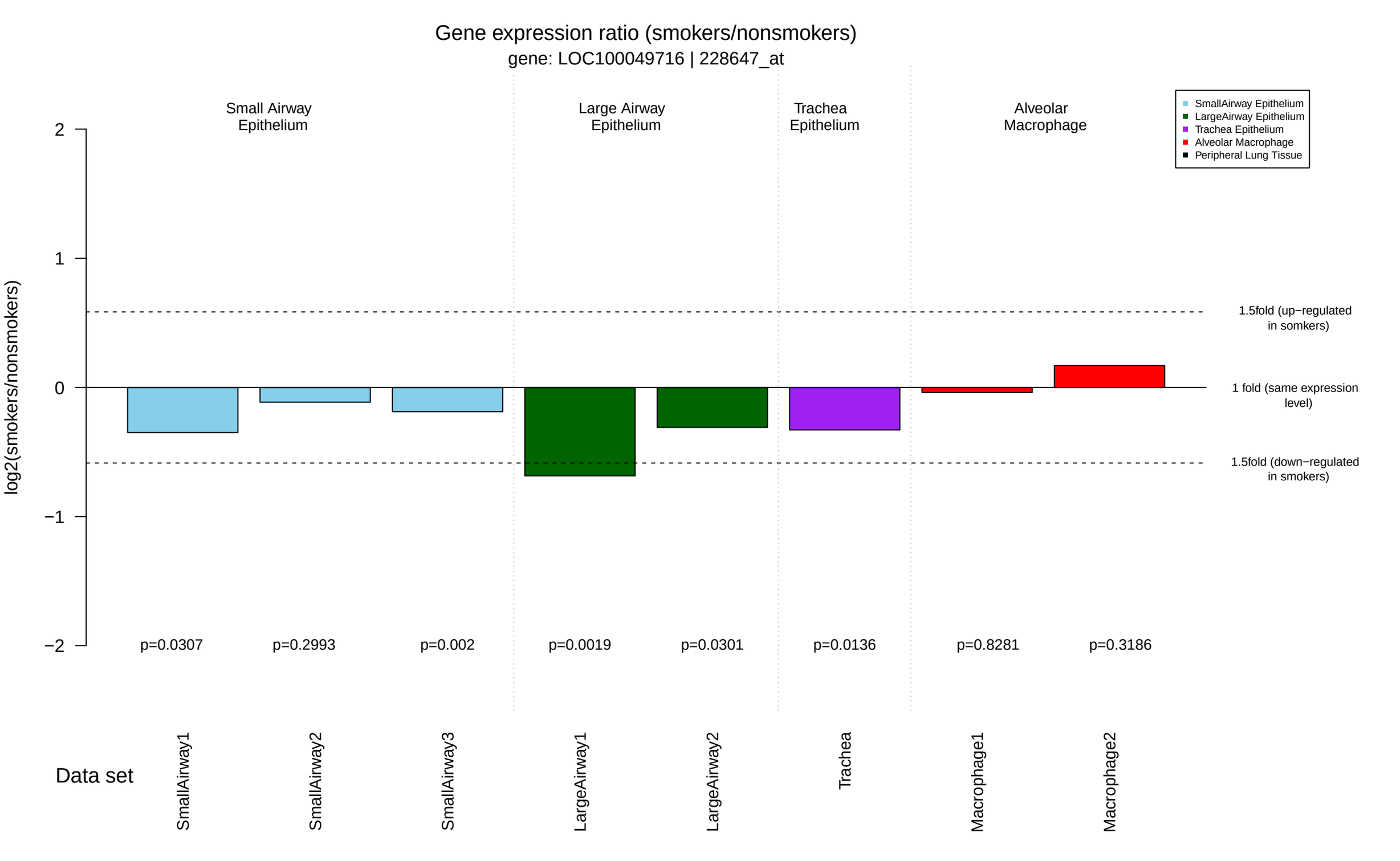
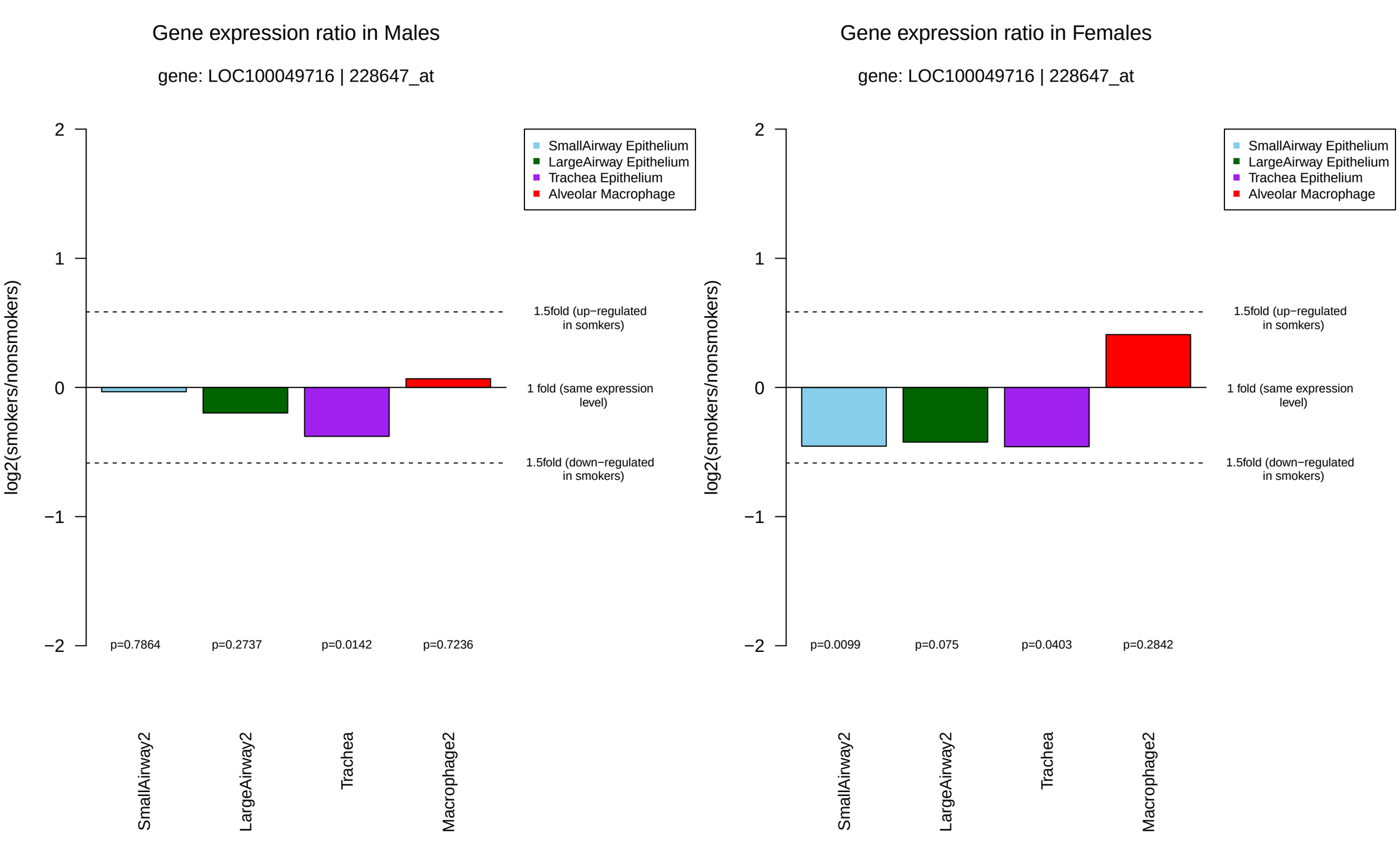
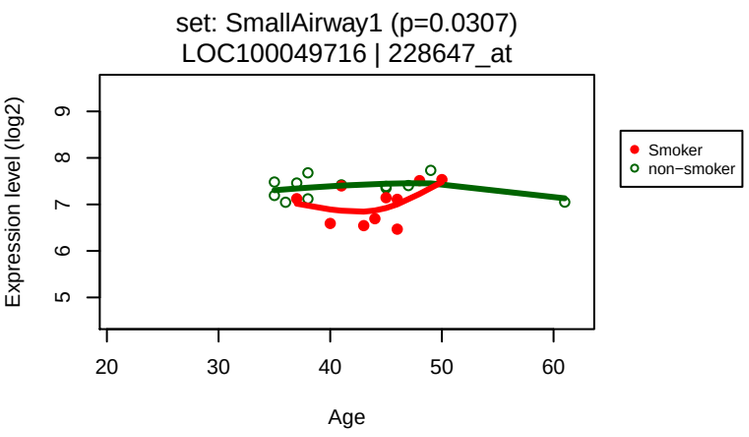
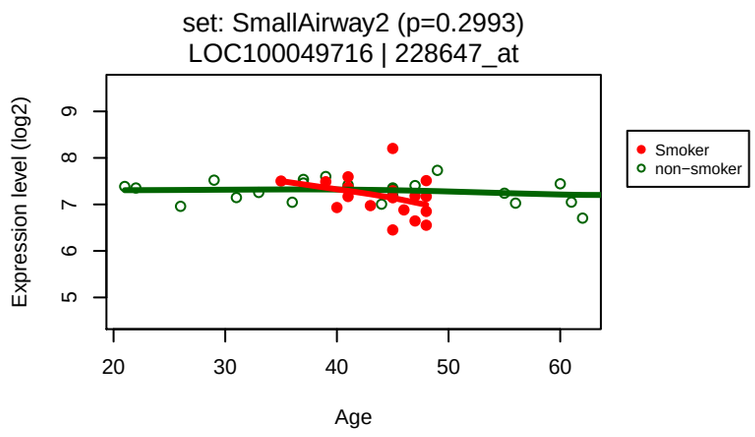
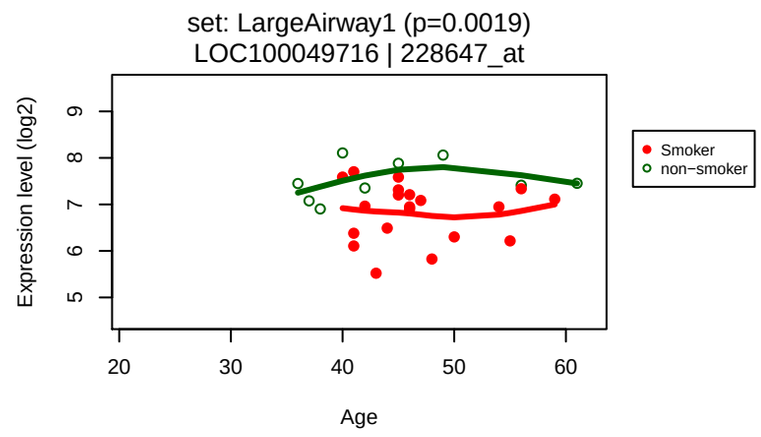

Gene expression of smokers and nonsmokers
gene: LOC100049716 | 228647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0307
p=0.2993
p=0.002
p=0.0019
p=0.0301
p=0.0136
p=0.8281
p=0.3186
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

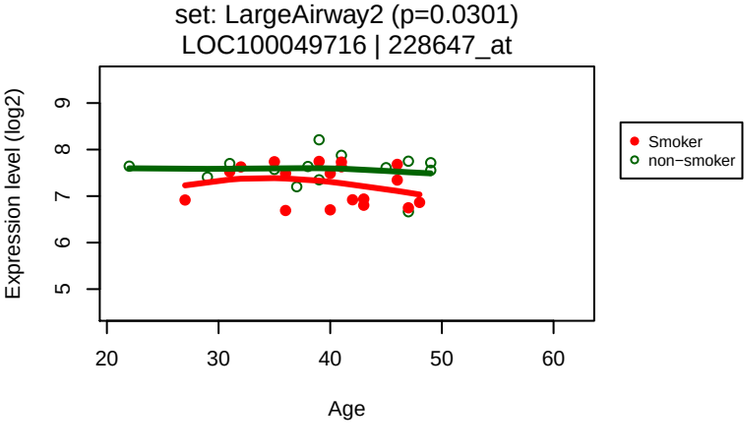

Gene expression ratio (smokers/nonsmokers)
gene: LOC100049716 | 228647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0307
p=0.2993
p=0.002
p=0.0019
p=0.0301
p=0.0136
p=0.8281
p=0.3186
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

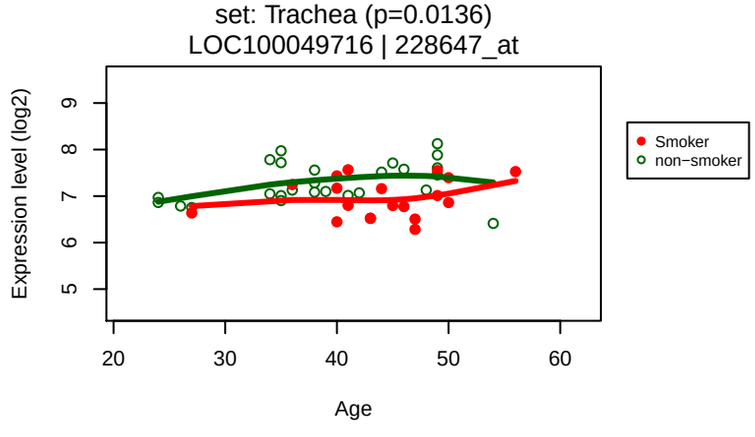

Gene expression ratio in Males
gene: LOC100049716 | 228647_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7864
p=0.2737
p=0.0142
p=0.7236
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC100049716 | 228647_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0099
p=0.075
p=0.0403
p=0.2842
SmallAirway2
LargeAirway2
Trachea
Macrophage2

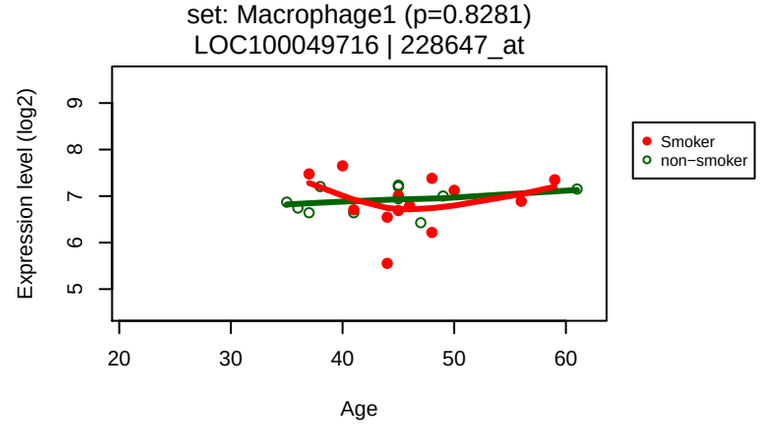

set: SmallAirway1 (p=0.0307)
LOC100049716 | 228647_at
Expression level (log2)
Age
Smoker
non−smoker

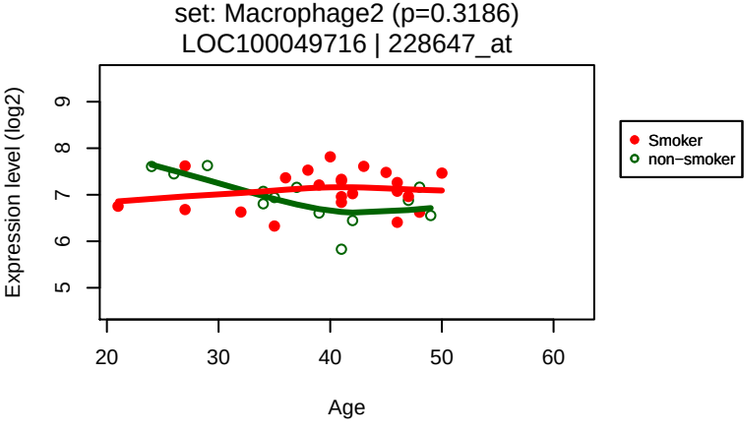

set: SmallAirway2 (p=0.2993)
LOC100049716 | 228647_at
Expression level (log2)
Age
Smoker
non−smoker

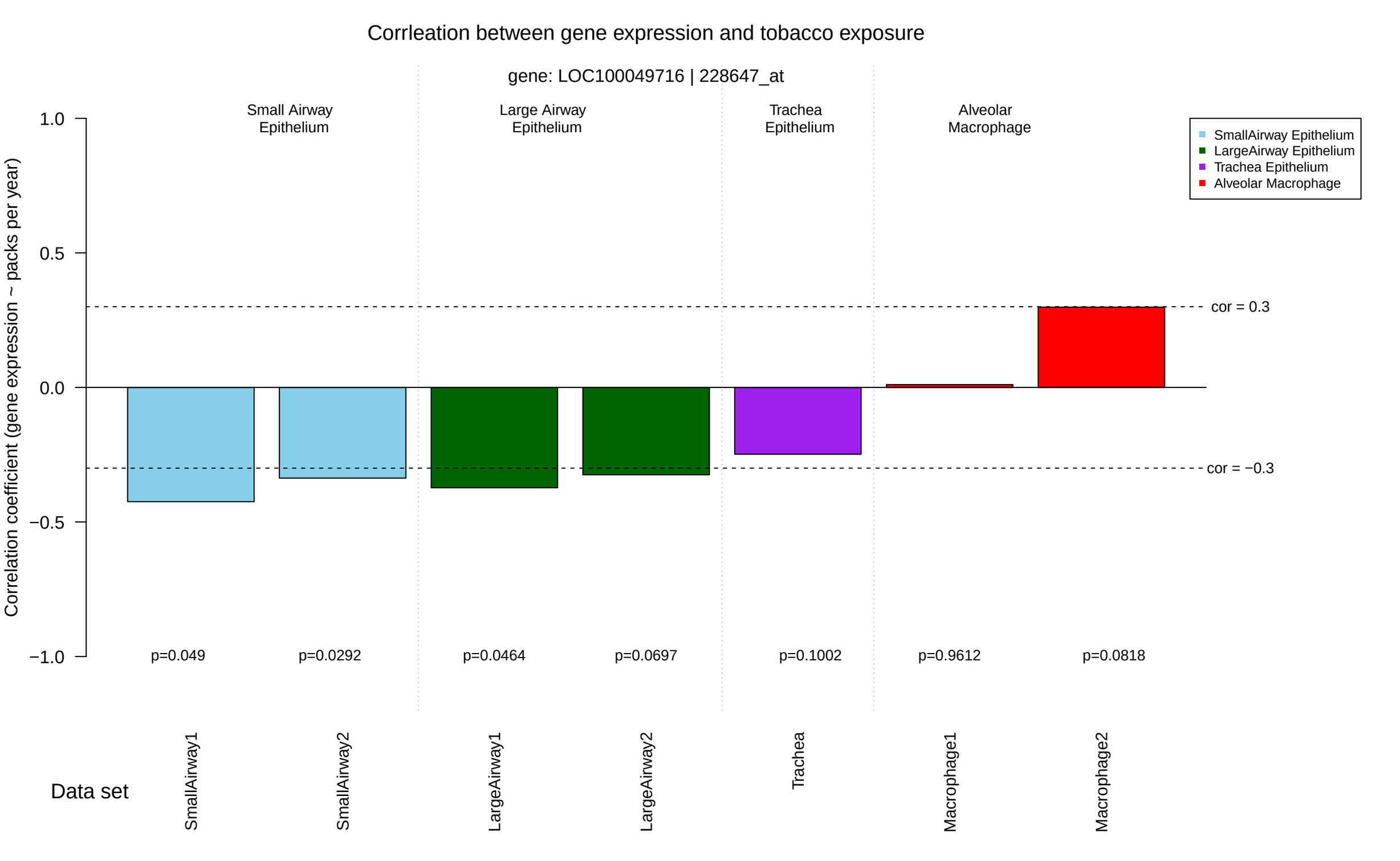

set: LargeAirway1 (p=0.0019)
LOC100049716 | 228647_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0301)
LOC100049716 | 228647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0136)
LOC100049716 | 228647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8281)
LOC100049716 | 228647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3186)
LOC100049716 | 228647_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC100049716 | 228647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.049
p=0.0292
p=0.0464
p=0.0697
p=0.1002
p=0.9612
p=0.0818
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2